

Lab - Azure Container Instances (ACI)

To Begin with the lab

Summary

This lab demonstrates how to deploy and run a Docker container using **Azure Container Instances (ACI)**. Unlike Azure Virtual Machines, ACI eliminates the need to manage servers or install Docker. You simply select a container image, configure networking, and Azure automatically provisions the infrastructure to run your container.

Understanding

Azure Container Instances (ACI) is a serverless container service that allows you to run Linux or Windows containers without managing virtual machines. Azure automatically provides the compute resources and Docker runtime. You only need to specify the container image, CPU/Memory requirements, networking, and ports.

ACI is best suited for simple containerized applications, testing, development environments, scheduled jobs, and short-lived workloads. For large-scale applications requiring orchestration, services like Azure Kubernetes Service (AKS) or Azure Container Apps are more appropriate.

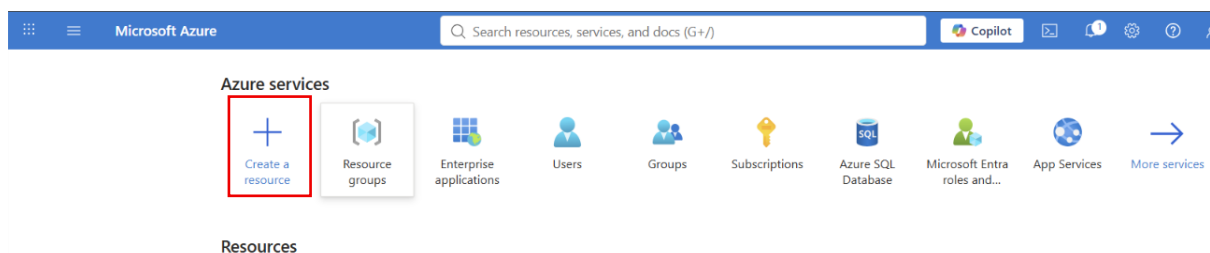
Example:

- Azure VM → You create and manage the VM, install Docker, and run containers.
- Azure Container Instance → Azure manages the infrastructure. You only deploy the container image.

Lab Steps

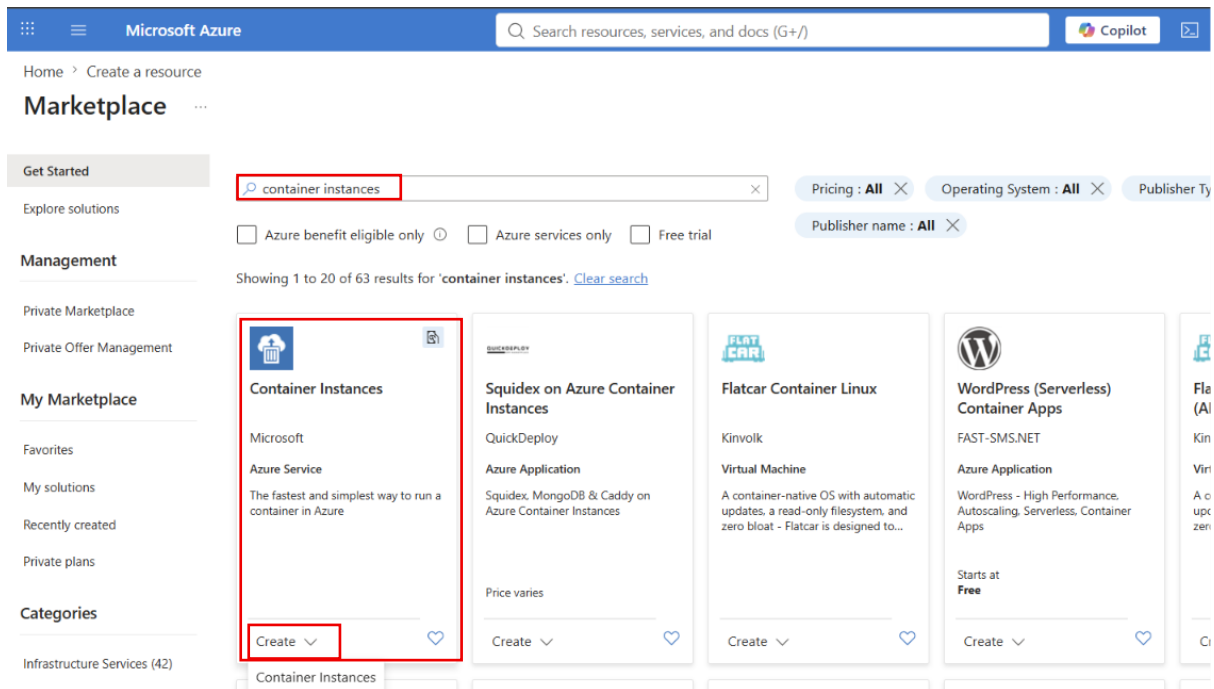
Step 1: Open Azure Portal

- Sign in to the Azure Portal.
- Open **All Resources** (optional).
- Click **Create a Resource**.



Step 2: Search for Container Instance

- Search for **Container Instances**.
- Select **Azure Container Instances**.
- Click **Create**.



Step 3: Configure Basic Settings

- Select the existing **Resource Group**.
- Enter a **Container Name** (Example: dockerapp).
- Choose the **Region** (Example: East US).

Home > Create a resource > Marketplace

Create container instance

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Azure Container Instances (ACI) allows you to quickly and easily run containers on Azure without managing servers or having to learn new tools. ACI offers per-second billing to minimize the cost of running containers on the cloud.
[Learn more about Azure Container Instances](#)

Project details

Select the subscription to manage deployed resources and costs. Use resource groups like folders to organize and manage all your resources.

Subscription *

Resource group *

[Create new](#)

Container details

Container name *

Region *

Step 4: Select Container Image

- Under **Image Source**, choose **Azure Container Registry**.
- Select your **Container Registry**.
- Select the Docker **Image**.
- Choose the appropriate **Image Tag**.
- Keep **OS Type** as **Linux**.

Microsoft Azure

Search resources, services, and docs (G+/)

Home > Create a resource > Marketplace

Create container instance

Region * (US) East US

Availability zones (Preview) None

SKU Standard

Image source * ☐ Quickstart images ☒ Azure Container Registry ☐ Other registry

Run with Azure Spot discount ☐

Spot containers are not available in the selected region. [Learn more](#)

Registry * regdevs12

If you do not see your Azure Container Registry, ensure you have been assigned the Reader Role for the Azure Container Registry or select an Azure Container Registry in a different subscription. [Learn more](#)

Image * dockerapp

Image tag * v1

OS type * ☒ Linux ☐ Windows

Size * 1 vcpu, 1.5 GiB memory, 0 gpus

[Change size](#)

Step 6: Configure Networking

- Go to the **Networking** tab.
- Set **Public IP** to **Enabled**.
- Add **Port 8080**.
- Keep protocol as **TCP**.

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Create container instance

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Choose between three networking options for your container instance:

- 'Public'** will create a public IP address for your container instance.
- 'Private'** will allow you to choose a new or existing virtual network for your container instance.
- 'None'** will not create either a public IP or virtual network. You will still be able to access your container logs using the command line.

Networking type ☒ Public ☐ Private ☐ None

DNS name label

DNS name label scope reuse Any reuse (unsecure)

Ports

Ports protocol

8080 TCP

Step 7: Monitoring

- Open the **Monitoring** tab.
- Leave **Container Insights** with the default settings.

Step 8: Advanced Settings

- Open the **Advanced** tab.

- Leave the default settings unless customization is required.

Step 9: Deploy the Container

- Click **Review + Create**.
- Validate the configuration.
- Click **Create**.
- Wait until deployment completes.

Create container instance ...

✓ Validation passed

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Basics

Subscription	Visual Studio Enterprise Subscription
Resource group	Webapp
Region	East US
Container name	dockerapp
SKU	Standard
Image type	Private
Image registry login server	regdeves12.azurecr.io
Image	regdeves12.azurecr.io/dockerapp:v1
Image registry user name	regdeves12
OS type	Linux
Memory (GiB)	1.5
Number of CPU cores	1
GPU type (preview)	None
GPU count	0

Networking

Networking type	Public
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Create

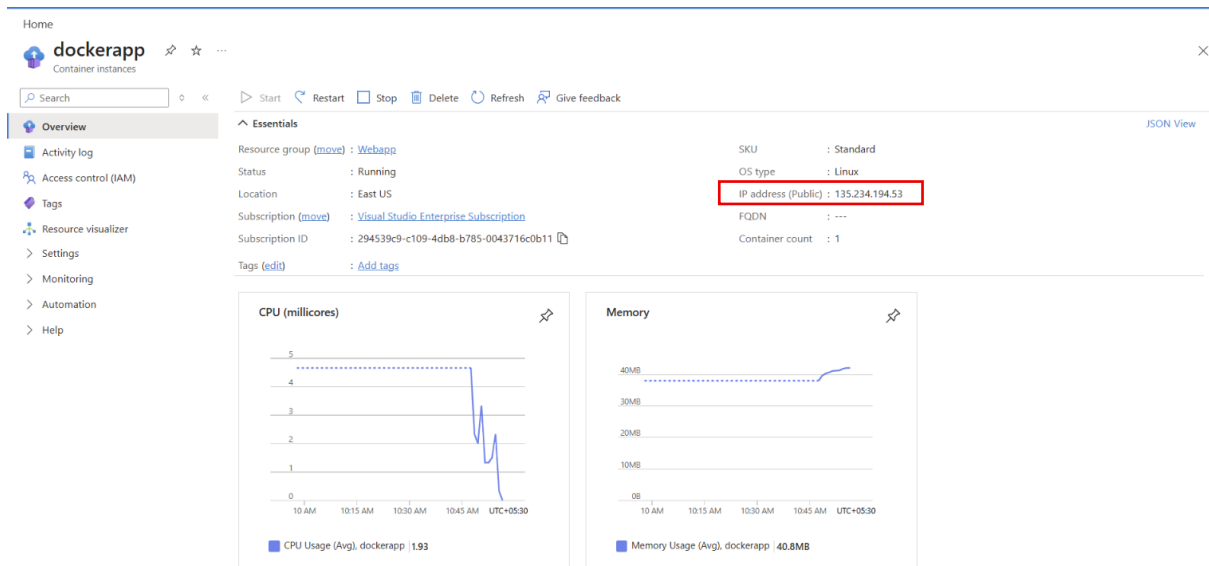
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Step 10: Open the Running Application

- Open the deployed **Container Instance**.
- Copy the **Public IP Address**.



- Open a browser.
- Browse to:
- <http://<Public-IP>:8080>



Welcome

Learn about [building Web apps with ASP.NET Core](#).

- Verify that the Docker application is displayed successfully.